



16 REPRODUCTION

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[VIEW EXAM QUESTIONS](#)

YOUR NOTES



16.1 TYPES OF REPRODUCTION

Asexual Reproduction

- Asexual reproduction does not involve sex cells or fertilisation
- Only one parent is required so there is no fusion of gametes and no mixing of genetic information
- As a result, the offspring are genetically identical to the parent and to each other (clones)
- Asexual reproduction is defined as a process resulting in genetically identical offspring from one parent

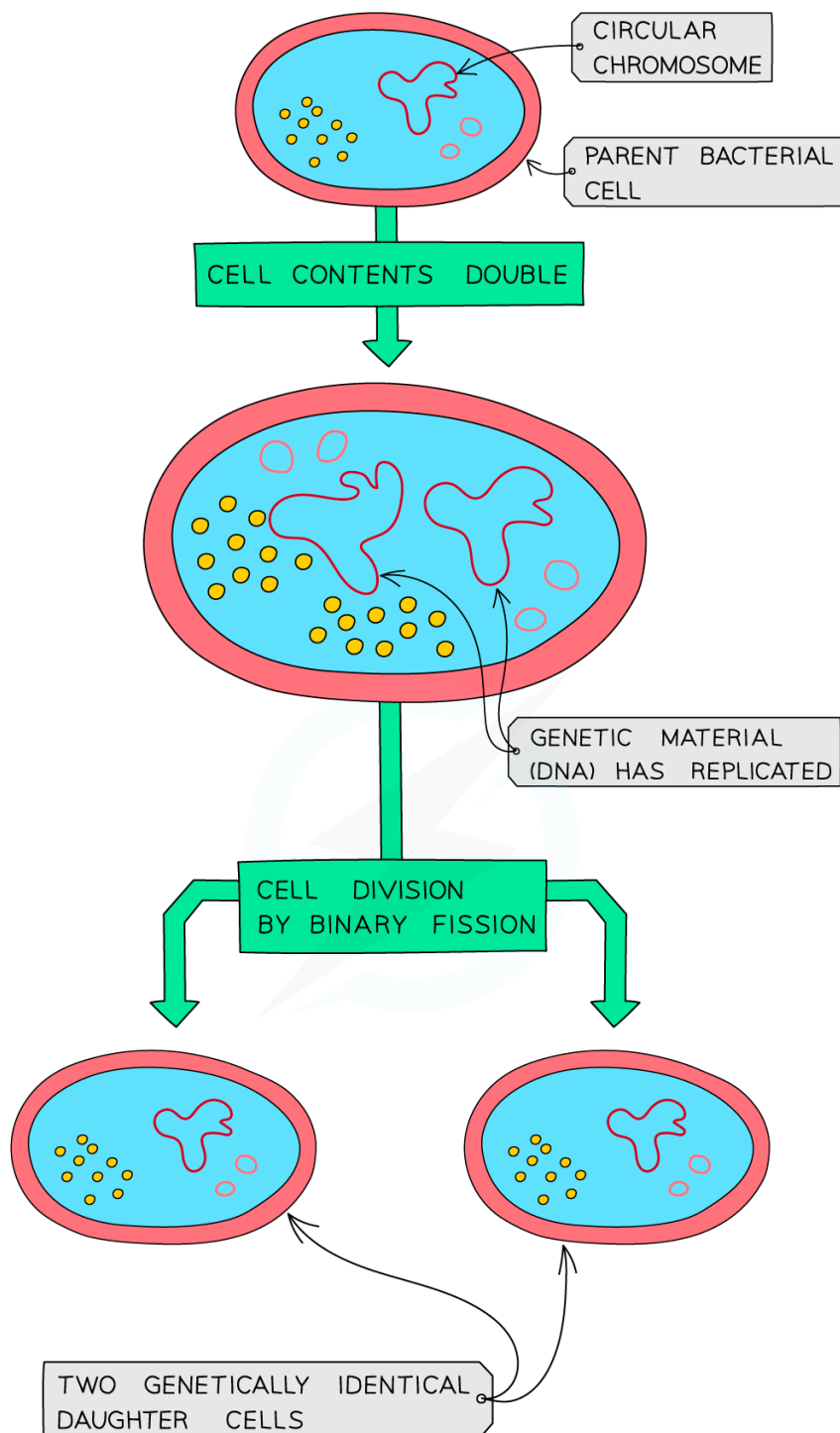
Bacteria produce exact genetic copies of themselves in a type of asexual reproduction called binary fission:



16 REPRODUCTION

16.1 TYPES OF REPRODUCTION cont...

YOUR NOTES



Bacterial binary fission



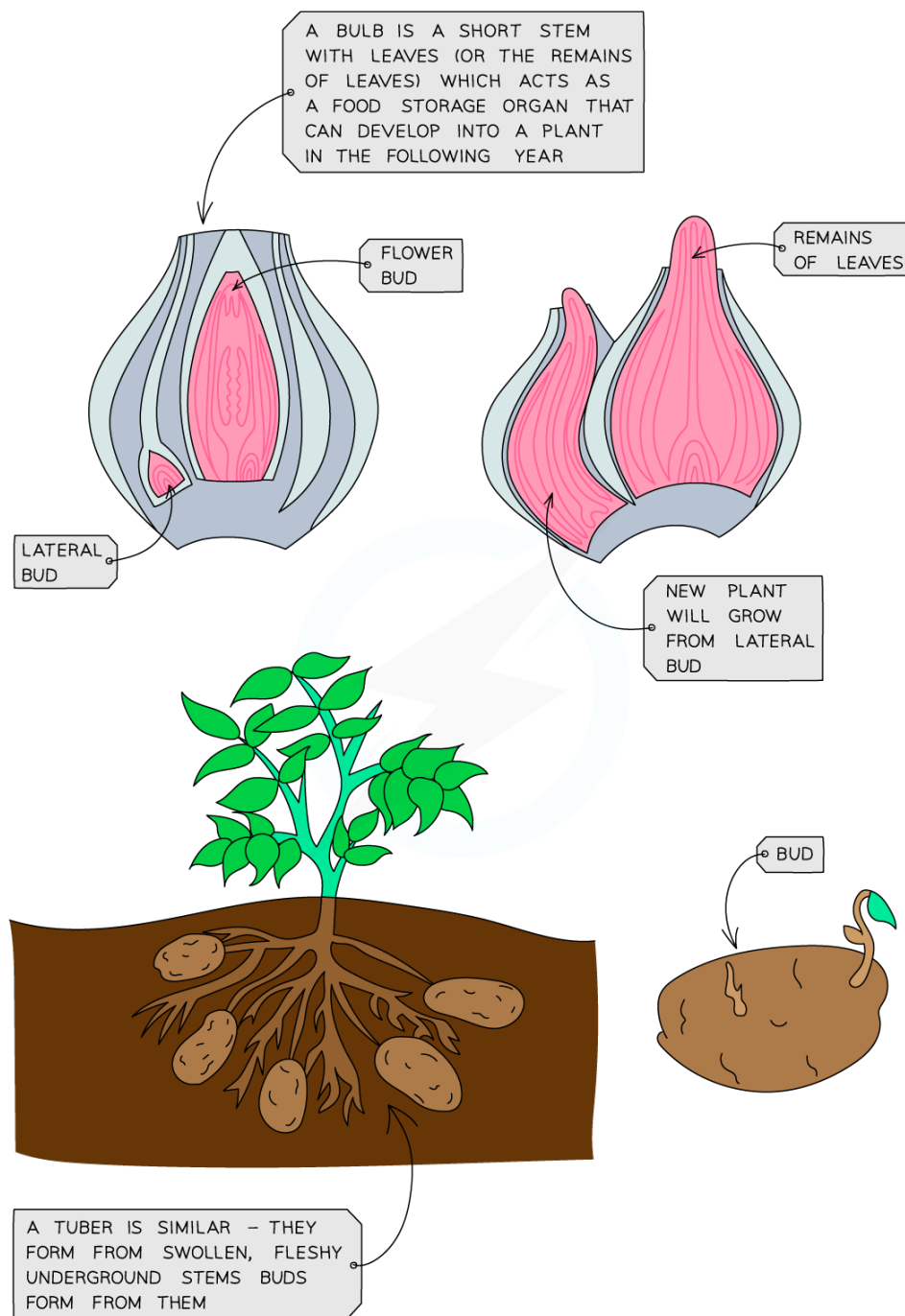
16 REPRODUCTION

16.1 TYPES OF REPRODUCTION cont...

YOUR NOTES



Plants can reproduce asexually using bulbs and tubers; these are food storage organs from which budding can occur, producing new plants which are genetically identical to the parent plant:



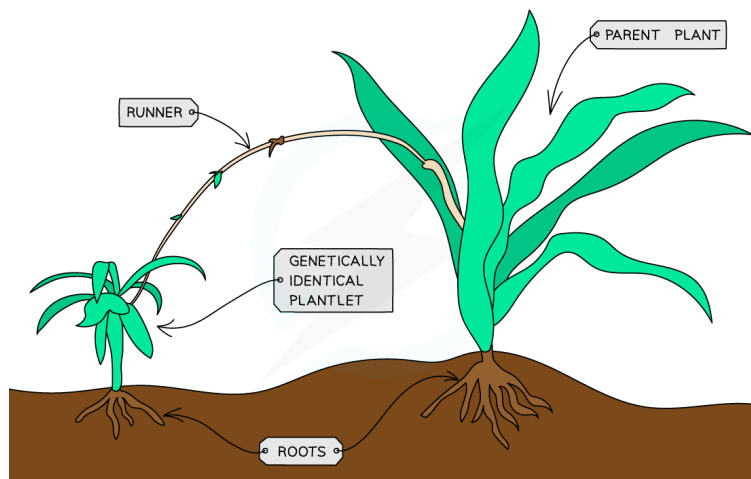
Asexual reproduction in plants using bulbs or tubers



16 REPRODUCTION

16.1 TYPES OF REPRODUCTION cont...

Some plants grow side shoots called runners that contain tiny plantlets on them (a good example of this are strawberry plants). These will grow roots and develop into separate plants, again being genetically identical to the parent plant:



A runner and plantlet

YOUR NOTES



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Advantages & Disadvantages of Asexual Reproduction

ADVANTAGES	DISADVANTAGES
POPULATION CAN BE INCREASED RAPIDLY WHEN CONDITIONS ARE RIGHT	LIMITED GENETIC VARIATION IN POPULATION – OFFSPRING ARE GENETICALLY IDENTICAL TO THEIR PARENTS
CAN EXPLOIT SUITABLE ENVIRONMENTS QUICKLY	POPULATION IS VULNERABLE TO CHANGES IN CONDITIONS AND MAY ONLY BE SUITED FOR ONE HABITAT
MORE TIME AND ENERGY EFFICIENT	DISEASE IS LIKELY TO AFFECT THE WHOLE POPULATION AS THERE IS NO GENETIC VARIATION
REPRODUCTION IS COMPLETED MUCH FASTER THAN SEXUAL REPRODUCTION	

- Specifically in **crop plants**, asexual reproduction can be advantageous as it means that a plant that has good characteristics (high yield, disease-resistant, hardy) can be made to reproduce asexually and the **entire crop will show the same characteristics**



16 REPRODUCTION

16.1 TYPES OF REPRODUCTION cont...

YOUR NOTES



Sexual Reproduction

- Sexual reproduction is a **process involving the fusion of the nuclei of two gametes (sex cells) to form a zygote (fertilised egg cell) and the production of offspring that are genetically different from each other**
- Fertilisation is defined as the **fusion of gamete nuclei**, and as each gamete comes from a different parent, there is variation in the offspring



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Gametes & Zygotes

- A **gamete** is a sex cell (in animals: sperm and ovum; in plants pollen nucleus and ovum)
- Gametes differ from normal cells as they contain **half the number of chromosomes** found in other body cells – we say they have a **haploid nucleus**
- This is because they only contain **one copy of each chromosome**, rather than the two copies found in other body cells
- In human beings, a normal body cell contains **46 chromosomes** but each gamete contains **23 chromosomes**
- When the male and female gametes fuse, they become a zygote (fertilised egg cell)
- This contains the full **46 chromosomes**, half of which came from the father and half from the mother – we say the zygote has a **diploid nucleus**

Advantages & Disadvantages of Sexual Reproduction

ADVANTAGES	DISADVANTAGES
INCREASES GENETIC VARIATION	TAKES TIME AND ENERGY TO FIND MATES
THE SPECIES CAN ADAPT TO NEW ENVIRONMENTS DUE TO VARIATION, GIVING THEM A SURVIVAL ADVANTAGE	DIFFICULT FOR ISOLATED MEMBERS OF THE SPECIES TO REPRODUCE
DISEASE IS LESS LIKELY TO AFFECT POPULATION (DUE TO VARIATION)	

- Most crop plants reproduce sexually and this is an advantage as it means **variation is increased** and a genetic variant may be produced which is better able to cope with weather changes, or produces significantly higher yield
- The disadvantage is that the variation may lead to offspring that are **less successful** than the parent plant at growing well or producing a good harvest



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16.2 SEXUAL REPRODUCTION IN PLANTS

YOUR NOTES



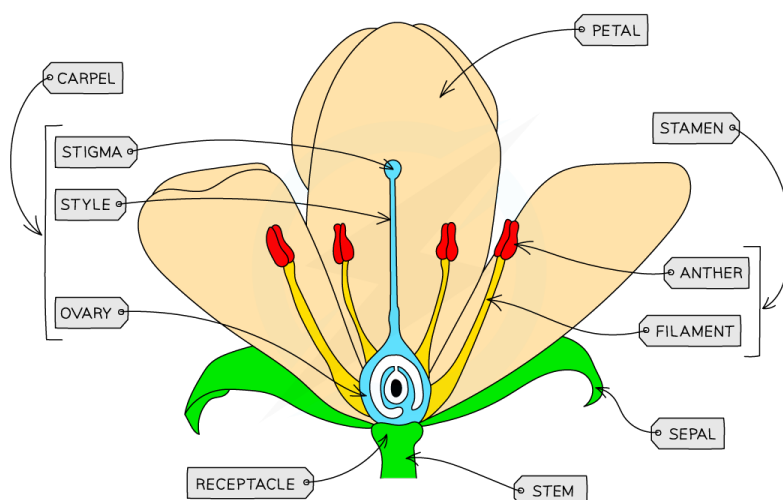
Flowers & Pollination

- Flowers are the **reproductive organ** of the plant
- They usually contain both male and female reproductive parts
- Plants produce **pollen** which contains a nucleus inside that is the **male gamete**
- Unlike the male gamete in humans (sperm), pollen is not capable of locomotion (moving from one place to another)
- This means plants have to have mechanisms in place to **transfer pollen from the anther to the stigma**
- This process is known as **pollination** and there are two main mechanisms by which it occurs: transferred by **insects** (or other animals like birds) or transferred by **wind**
- The structure of insect and wind-pollinated flowers are slightly different as each is adapted for their specific function

Parts of a Flower

STRUCTURE	DESCRIPTION
SEPAL	PROTECTS UNOPENED FLOWER
PETALS	BRIGHTLY COLOURED IN INSECT – POLLINATED FLOWERS TO ATTRACT INSECTS
ANTHER	PRODUCES AND RELEASES THE MALE SEX CELL (POLLEN GRAIN)
STIGMA	TOP OF THE FEMALE PART OF THE FLOWER WHICH COLLECTS POLLEN GRAINS
OVARY	PRODUCES THE FEMALE SEX CELL (OVUM)
OVULE	CONTAINS THE FEMALE SEX CELLS (FOUND INSIDE THE OVARY)

General flower structure





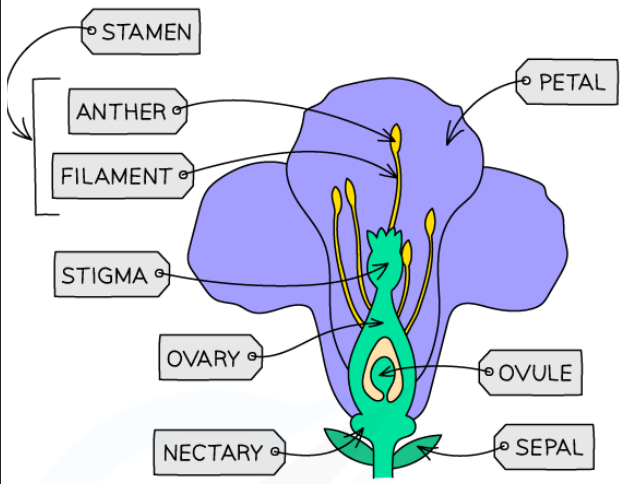
16 REPRODUCTION

16.2 SEXUAL REPRODUCTION IN PLANTS cont...

YOUR NOTES



Adaptations for pollination

FEATURE	INSECT-POLLINATED
	 FEATURES OF AN INSECT-POLLINATED FLOWER
PETALS	LARGE AND BRIGHTLY COLOURED TO ATTRACT INSECTS
SCENT AND NECTAR	PRESENT – ENTICES INSECTS TO VISIT THE FLOWER AND PUSH PAST STAMEN TO GET TO NECTAR
NUMBER OF POLLEN GRAINS	MODERATE – INSECTS TRANSFER POLLEN GRAINS EFFICIENTLY WITH A HIGH CHANCE OF SUCCESSFUL POLLINATION
POLLEN GRAINS	LARGER, STICKY AND / OR SPIKY TO ATTACH TO INSECTS AND BE CARRIED AWAY
ANTHERS	INSIDE FLOWER, STIFF AND FIRMLY ATTACHED TO BRUSH AGAINST INSECTS
STIGMA	INSIDE FLOWER, STICKY SO POLLEN GRAINS STICK TO IT WHEN AN INSECT BRUSHES PAST

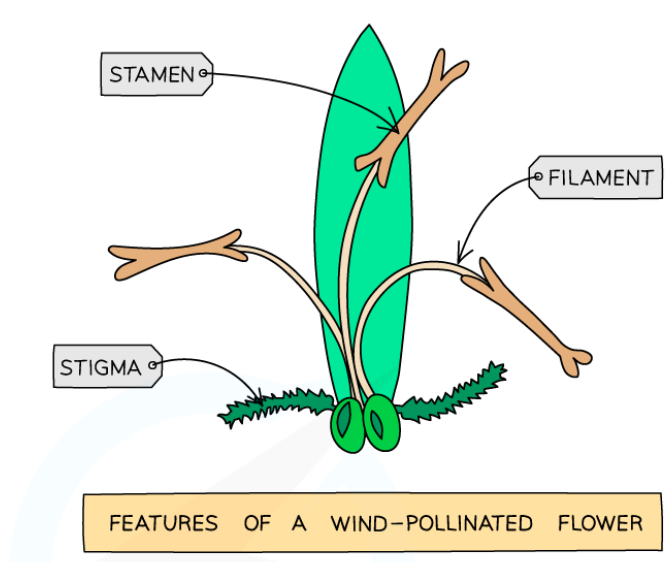


16 REPRODUCTION

16.2 SEXUAL REPRODUCTION IN PLANTS cont...

YOUR NOTES



FEATURE	WIND-POLLINATED
	
PETALS	SMALL AND DULL, OFTEN GREEN OR BROWN IN COLOUR
SCENT AND NECTAR	ABSENT – NO NEED TO WASTE ENERGY PRODUCING THESE AS NO NEED TO ATTRACT INSECTS
NUMBER OF POLLEN GRAINS	LARGE AMOUNTS – MOST POLLEN GRAINS ARE NOT TRANSFERRED TO ANOTHER FLOWER SO THE MORE PRODUCED, THE BETTER THE CHANCE OF SOME SUCCESSFUL POLLINATION OCCURRING
POLLEN GRAINS	SMOOTH, SMALL AND LIGHT SO THEY ARE EASILY BLOWN BY THE WIND
ANTHERS	OUTSIDE FLOWER, SWINGING LOOSE ON LONG FILAMENTS TO RELEASE POLLEN GRAINS EASILY
STIGMA	OUTSIDE FLOWER, FEATHERY TO CATCH DRIFTING POLLEN GRAINS

- The pollen produced by insect and wind-pollinated flowers is also different:
 - Insect-pollinated flowers produce **smaller amounts** of **larger, heavier** pollen grains that often contain **spikes or hooks** on the outside so they are better able to stick to insects
 - Wind-pollinated flowers produce **large amounts** of **small, lightweight** pollen grains that are usually **smooth**

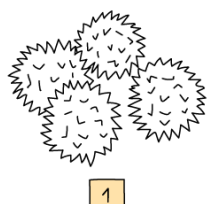


16 REPRODUCTION

16.2 SEXUAL REPRODUCTION IN PLANTS cont...

Here is an example of a multiple-choice question asking students to use their knowledge to identify types of pollen grain:

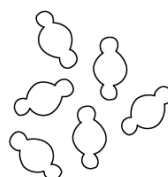
THE DIAGRAMS SHOW POLLEN GRAINS FROM THREE DIFFERENT SPECIES OF PLANT AS THEY APPEAR UNDER THE MICROSCOPE. THE DIAGRAMS ARE ALL TO THE SAME SCALE.



1



2



3

WHICH POLLEN GRAINS ARE INVOLVED IN INSECT-POLLINATION?

- A. 1 AND 2 B. 1 ONLY C. 2 AND 3 D. 3 ONLY

A multiple-choice question asking students to use their knowledge to identify types of pollen grain



EXTENDED ONLY

Self & Cross-Pollination

- Cross-pollination occurs when the **pollen from one plant** is transferred to the stigma of **another plant of the same species**
- This is the way most plants carry out pollination as it **improves genetic variation**
- Occasionally, the pollen from a flower can **land on its own stigma** or on the **stigma of another flower on the same plant** – this is known as self-pollination
- Self-pollination **reduces genetic variety of the offspring** as all the gametes come from the same parent (and are therefore genetically identical)
- Lack of variation in the offspring is a disadvantage if environmental conditions change, as it is **less likely that any offspring will have adaptations that suit the new conditions** well
- On the other hand, cross-pollination relies completely on the presence of pollinators and this can be a problem if those **pollinators** are **missing** (eg the reduction in **bee** numbers is of great importance to humans as bees pollinate a large number of food crops) – this doesn't apply to wind-pollinated plants

YOUR NOTES





16 REPRODUCTION

16.2 SEXUAL REPRODUCTION IN PLANTS cont...

YOUR NOTES



Fertilisation: Basics

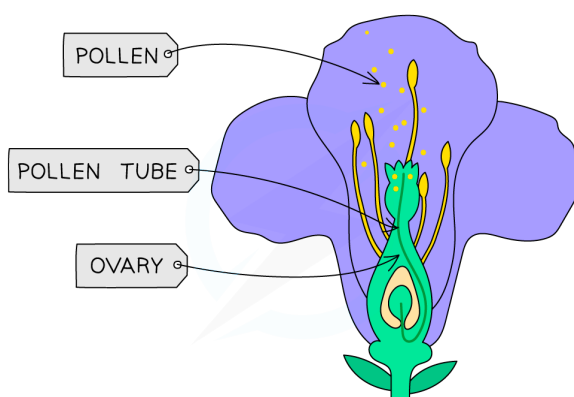
- Fertilisation occurs when a **pollen nucleus fuses with an ovum nucleus** in the ovule



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Fertilisation

- As the pollen has no 'tail' to swim to the ovary of a plant, in order to reach the 'female' nucleus in the ovary it has to grow a **pollen tube**
- This only happens if the pollen grain has landed on the right kind of stigma (i.e. of the same species as the flower the pollen came from)
- The nucleus inside the pollen grain **slips down the tube** as it grows down the style towards the ovary
- The ovary contains one or more **ovules** which each contain an ovum with a female nucleus that a male pollen nucleus can fuse with
- Once the nuclei (pl) have joined together, that ovule has been fertilised and a **zygote** has been formed
- The zygote will start to divide and eventually form a **seed** within the ovule
- As different plants have different numbers of ovules, this explains why different fruits (which develop from the ovary) have different numbers of seeds (which develop from the ovules)



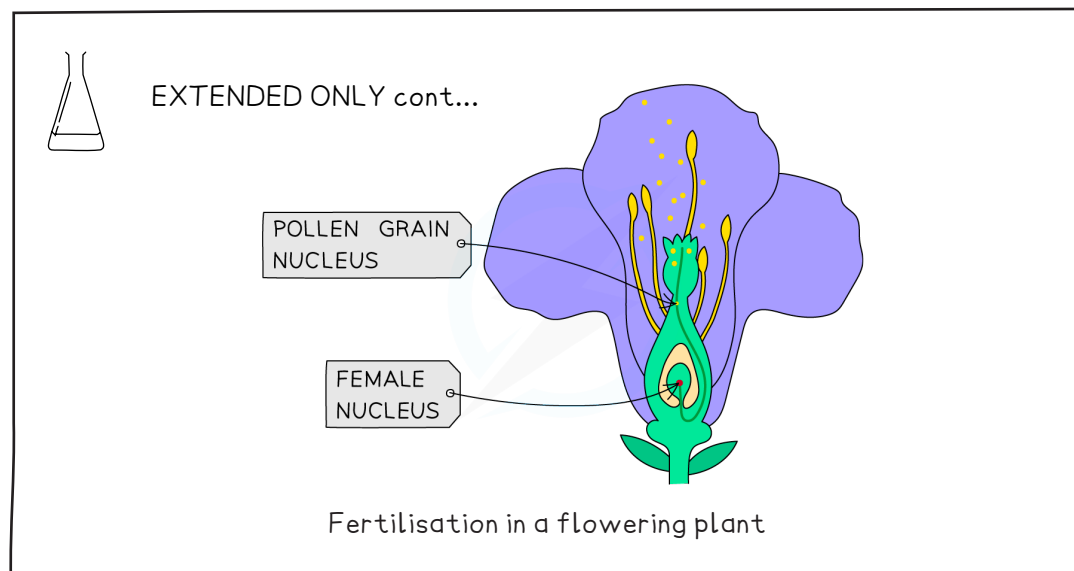
Growth of a pollen tube



16 REPRODUCTION

16.2 SEXUAL REPRODUCTION IN PLANTS cont...

YOUR NOTES



EXAM TIP

Students often get confused between pollination and fertilisation in plants, but they are not the same thing.

Think of pollination as the plant's equivalent to human sexual intercourse – after sex, the male sex cells (sperm) have been deposited into the female.

But, for fertilisation to occur, the **nucleus from a male sperm cell has to fuse with the nucleus of a female sex cell** (egg) and the sperm has to travel to find the egg before this happens. It's exactly the same in plants!

16.3 GERMINATION

Factors Affecting Germination

- **Germination** is the **start of growth in the seed**
- Three factors are required for successful germination:
 - **Water** – allows the seed to swell up and the enzymes in the embryo to start working so that growth can occur
 - **Oxygen** – so that energy can be released for germination
 - **Warmth** – germination improves as temperature rises (up to a maximum) as the reactions which take place are controlled by enzymes
- As carbon dioxide is not necessary for germination but also does not inhibit it, it makes no difference whether it is present or not



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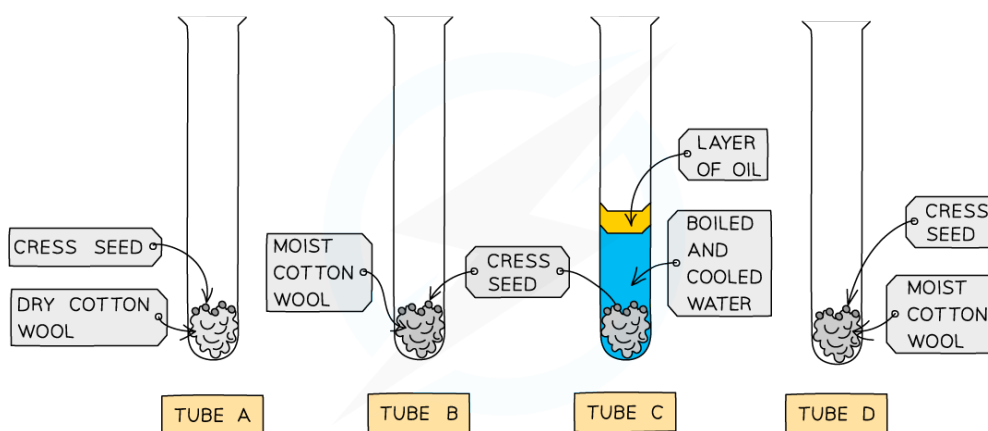
16.3 GERMINATION cont...

YOUR NOTES



Investigating Germination

- Set up 4 boiling tubes each containing 10 cress seeds on cotton wool
- Set each test tube as shown in diagram below
- Leave tubes in set environment for a period of time: A, B and C incubated at 20°C; D placed in a fridge at 4°C
- Compare results and see which tube has the greatest number of germinated seeds



Conditions required for germination

TEST TUBE	FACTOR BEING TESTED	SEEDS GERMINATED
A	WATER / MOISTURE	NO
B	CONTROL (ALL FACTORS PRESENT)	YES
C	OXYGEN	NO
D	WARM TEMPERATURE	NO



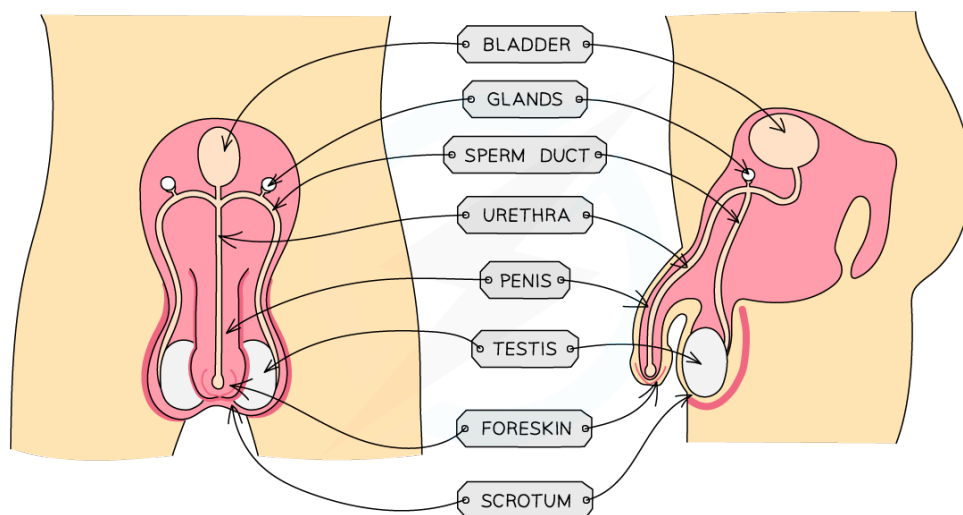
16 REPRODUCTION

16.4 SEXUAL REPRODUCTION IN HUMANS

YOUR NOTES



The Male Reproductive System



Male reproductive structures

STRUCTURE	FUNCTION
PROSTATE GLAND	PRODUCES FLUID CALLED SEMEN THAT PROVIDE SPERM CELLS WITH NUTRIENTS
SPERM DUCT	SPERM PASSES THROUGH THE SPERM DUCT TO BE MIXED WITH FLUIDS PRODUCED BY THE GLANDS BEFORE BEING PASSED INTO THE URETHRA FOR EJACULATION
URETHRA	TUBE RUNNING DOWN THE CENTRE OF THE PENIS THAT CAN CARRY OUT URINE OR SEMEN, A RING OF MUSCLE IN THE URETHRA PREVENTS THE URINE AND SEMEN FROM MIXING
TESTIS	CONTAINED IN A BAG OF SKIN (SCROTUM) AND PRODUCES SPERM (MALE GAMETE) AND TESTOSTERONE (HORMONE)
SCROTUM	SAC SUPPORTING THE TESTES OUTSIDE THE BODY TO ENSURE SPERM ARE KEPT AT TEMPERATURE SLIGHTLY LOWER THAN BODY TEMPERATURE
PENIS	PASSES URINE OUT OF THE BODY FROM THE BLADDER AND ALLOWS SEMEN TO PASS INTO THE VAGINA OF A WOMAN DURING SEXUAL INTERCOURSE



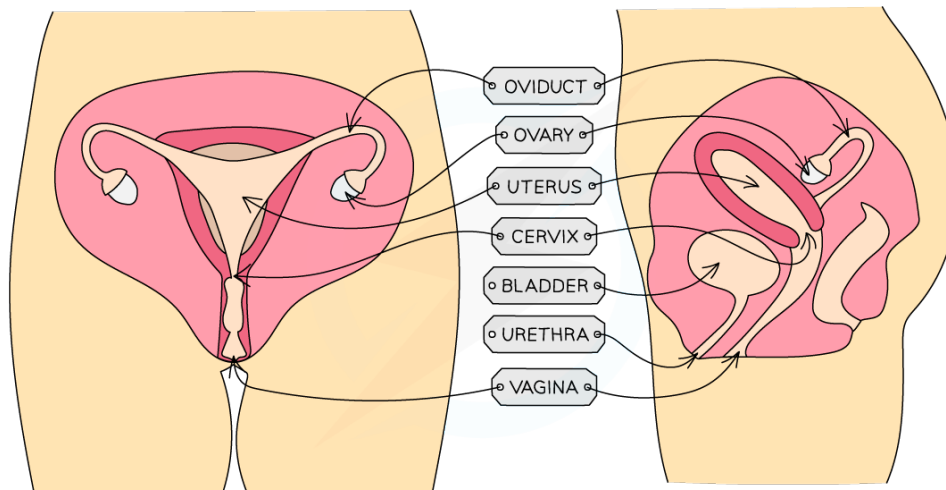
16 REPRODUCTION

16.4 SEXUAL REPRODUCTION IN HUMANS cont...

YOUR NOTES



The Female Reproductive System



Female reproductive structures

STRUCTURE	FUNCTION
OVIDUCT	CONNECTS THE OVARY TO THE UTERUS AND IS LINED WITH CILIATED CELLS TO PUSH THE RELEASED OVUM DOWN IT. FERTILISATION OCCURS HERE
OVARY	CONTAINS OVA (FEMALE GAMETES) WHICH WILL MATURE AND DEVELOP WHEN HORMONES ARE RELEASED
UTERUS	MUSCULAR BAG WITH A SOFT LINING WHERE THE FERTILISED EGG (ZYGOTE) WILL BE IMPLANTED TO DEVELOP INTO A FOETUS
CERVIX	RING OF MUSCLE AT THE LOWER END OF THE UTERUS TO KEEP THE DEVELOPING FOETUS IN PLACE DURING PREGNANCY
VAGINA	MUSCULAR TUBE THAT LEADS TO THE INSIDE OF THE WOMAN'S BODY, WHERE THE MALE'S PENIS WILL ENTER DURING SEXUAL INTERCOURSE AND SPERM ARE DEPOSITED



16 REPRODUCTION

16.4 SEXUAL REPRODUCTION IN HUMANS cont...

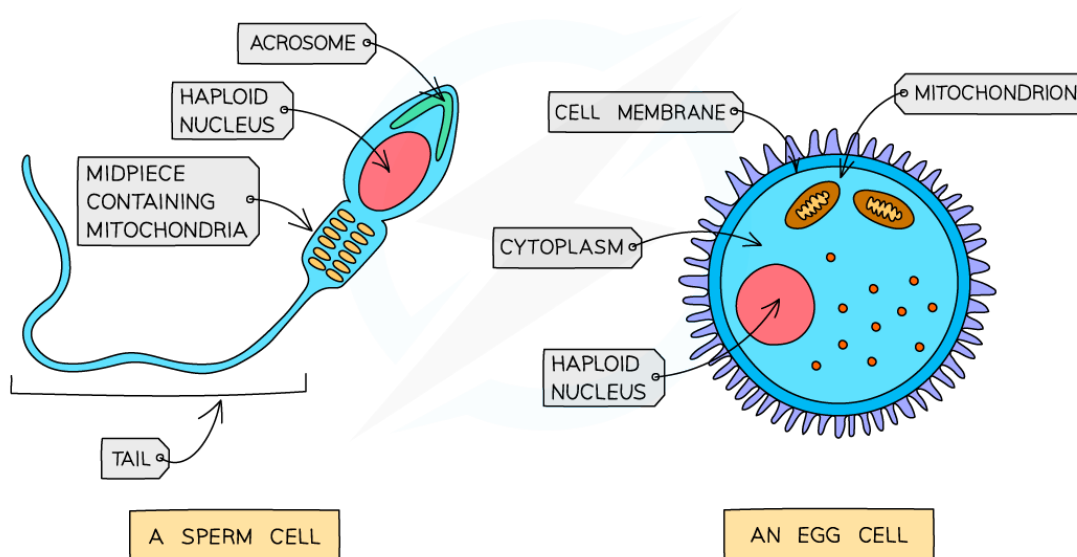
YOUR NOTES



Gametes & Fertilisation

- Fertilisation is the **fusion of the nuclei from a male gamete (sperm cell) and a female gamete (egg cell)**
- It occurs in the **oviducts**
- Gametes have **adaptations** to increase the chances of fertilisation and successful development of an embryo

SPERM CELL	EGG CELL
HAS A FLAGELLUM (TAIL)	CYTOPLASM CONTAINING A STORE OF ENERGY
CONTAINS ENZYMES IN THE HEAD REGION (ACROSOME)	JELLY-LIKE COATING THAT CHANGES AFTER FERTILISATION



Comparing sperm and egg cells



16 REPRODUCTION

16.4 SEXUAL REPRODUCTION IN HUMANS cont...

YOUR NOTES



EXTENDED ONLY

Adaptations of Gametes Explained

GAMETE	ADAPTIVE FEATURE	REASON
SPERM	HAS A FLAGELLUM (TAIL)	ENABLES IT TO SWIM TO THE EGG
	CONTAINS ENZYMES IN THE HEAD REGION (ACROSOME)	TO DIGEST THROUGH THE JELLY COAT AND CELL MEMBRANE OF AN EGG CELL WHEN IT MEETS ONE
	CONTAINS MANY MITOCHONDRIA	PROVIDE ENERGY FROM RESPIRATION SO THAT THE FLAGELLUM CAN MOVE BACK AND FORTH FOR LOCOMOTION
EGG	CYTOPLASM CONTAINING A STORE OF ENERGY	PROVIDES ENERGY FOR THE DIVIDING ZYGOTE AFTER FERTILISATION
	JELLY LIKE COATING THAT CHANGES AFTER FERTILISATION	FORMS AN IMPENETRABLE BARRIER AFTER FERTILISATION TO PREVENT OTHER SPERM NUCLEI ENTERING THE EGG CELL

Comparison of Male & Female Gametes

	SPERM	EGG
SIZE	VERY SMALL (45 μm)	LARGE (0.2mm)
STRUCTURE	HEAD REGION AND FLAGELLUM, MANY STRUCTURAL ADAPTATIONS	ROUND CELL WITH FEW STRUCTURAL ADAPTATIONS, COVERED IN A JELLY COATING
MOTILITY	CAPABLE OF LOCOMOTION	NOT CAPABLE OF LOCOMOTION
NUMBERS	PRODUCED EVERY DAY IN HUGE NUMBERS (AROUND 100 MILLION PER DAY)	THOUSANDS OF IMMATURE EGGS IN EACH OVARY, BUT ONLY ONE RELEASED EACH MONTH



16 REPRODUCTION

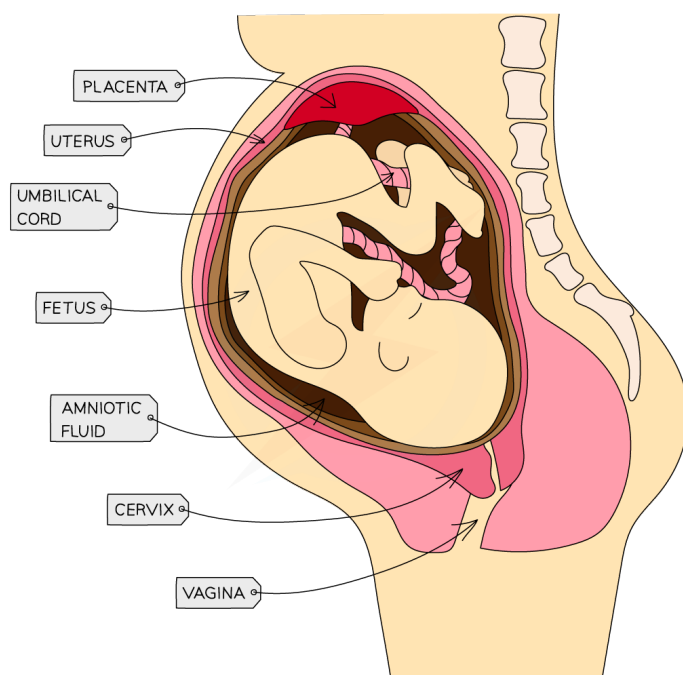
16.5 PREGNANCY & BIRTH

YOUR NOTES



Growth & Development of the Fetus

- After fertilisation in the **oviduct**, the zygote travels towards the **uterus**
- This takes about 3 days, during which time the zygote will divide several times to form a ball of cells known as an **embryo**
- In the uterus, the embryo embeds itself in the thick lining (**implantation**) and continues to grow and develop
- The gestation period for humans is **9 months**
- Major **development of organs** takes place within the **first 12 weeks**, during which time the embryo gets nutrients from the mother by **diffusion through the uterus lining**
- After this point the organs are all in place, the **placenta** has formed and the embryo is now called a **fetus**
- The remaining gestation time is used by the fetus to **grow bigger in size**



The fetus in the uterus

- The fetus is surrounded by an **amniotic sac** which contains **amniotic fluid** (made from the mother's blood plasma)
- This protects the fetus during development by **cushioning it from bumps** to the mother's abdomen
- The **umbilical cord** joins the fetus's blood supply to the **placenta** for exchange of nutrients and removal of waste products



16 REPRODUCTION

16.5 PREGNANCY & BIRTH cont...

YOUR NOTES



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The Placenta

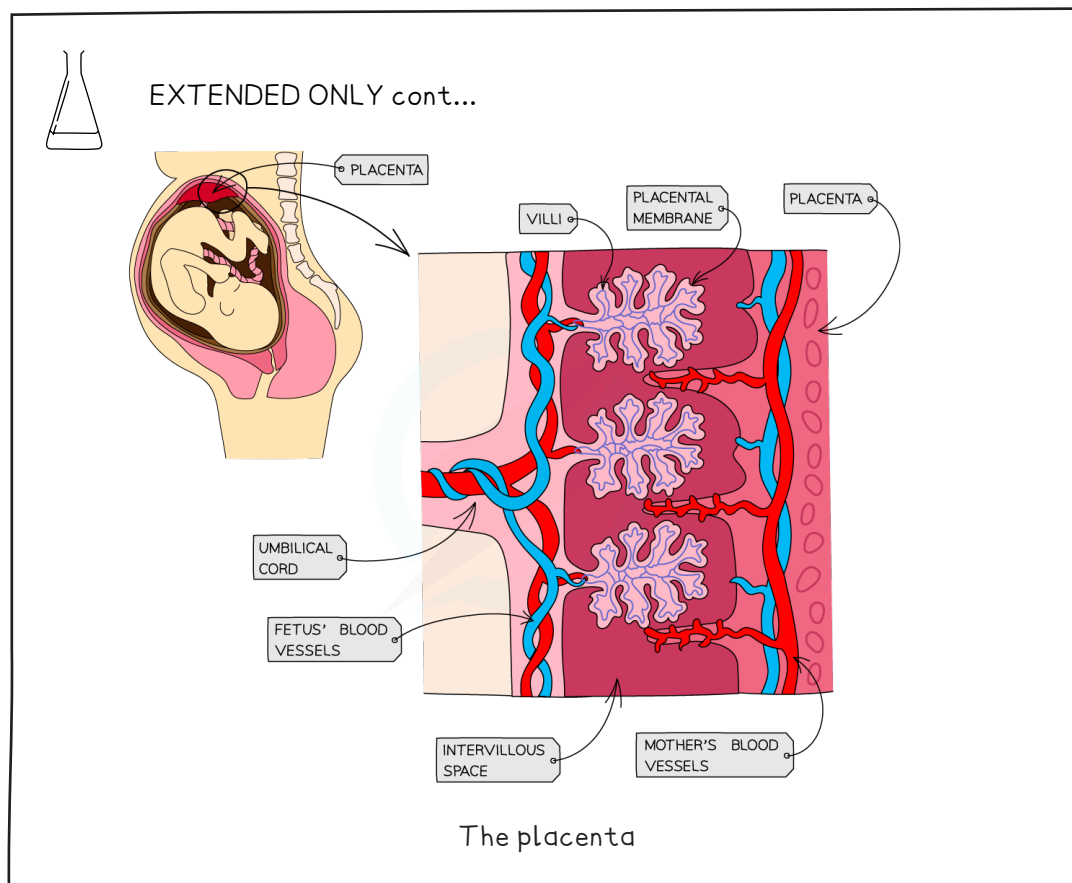
- During the gestation period the fetus develops and grows by gaining the **glucose, amino acids, fats, water** and **oxygen** it needs from the mother's blood
- The bloods run opposite each other, never mixing, in the **placenta**
- The fetus's blood connects to and from the placenta by the **umbilical cord**
- The mother's blood also absorbs the **waste** from the fetus's blood in the placenta; substances like **carbon dioxide** and **urea** are removed from the fetus's blood so that they do not build up to dangerous levels
- Movement of all molecules across the placenta occurs by **diffusion** due to **difference in concentration gradients**
- The placenta is adapted for this diffusion by having a **large surface area** and a **thin wall** for efficient diffusion
- The placenta acts as a **barrier** to prevent toxins and pathogens getting into the fetus's blood
- Not all **toxin molecules** or **pathogenic organisms** (such as **viruses**, eg **rubella**) are stopped from passing through the placenta (this usually depends on the size of the molecule)
- This is why pregnant women are advised **not to smoke during pregnancy** as molecules like **nicotine** can pass across the placenta
- After the baby has been born, the **umbilical cord is cut** – this does not hurt as there are no nerves in it, just two blood vessels
- It is tied off to prevent bleeding and shrivels up and falls off after a few days leaving the belly button behind
- The placenta detaches from the uterus wall shortly after birth and is pushed out due to contractions in the muscular wall of the uterus – known as the **afterbirth**



16 REPRODUCTION

16.5 PREGNANCY & BIRTH cont...

YOUR NOTES



EXAM TIP

It is worth learning at least **two specific substances** that move in either direction across the placenta

This is a common exam question and non-specific answers such as 'waste products' and 'nutrients' will not get any marks!

Stages of Birth

- Amniotic sac breaks
- Muscles in the uterus wall contract
- Cervix dilates (gets wider)
- Baby passes out through the vagina
- Umbilical cord is tied and cut
- Afterbirth is delivered



16 REPRODUCTION

16.5 PREGNANCY & BIRTH cont...

YOUR NOTES



Antenatal Care

- Antenatal (**before birth**), care is the name given to the care and advice given to expectant mothers along with checks on fetal growth and development
- Whilst pregnant, expectant mothers are given advice on:
 - **diet** including the need to take **folic acid** to prevent developmental issues with the fetus and the importance of a **balanced diet**
 - **exercise** to stay fit
 - health precautions such as avoiding infections, **tobacco**, **alcohol** and other **drugs**



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Breastfeeding

- During pregnancy the mammary glands enlarge and become prepared to secrete milk
- Shortly after birth, the mother will be stimulated to release milk due to the sucking action of the baby at the breast
- Some mothers struggle to breastfeed successfully and so may feed the baby using formula milk in a bottle

TYPE OF FEEDING	ADVANTAGES	DISADVANTAGES
BREASTFEEDING	CONTAINS EXACTLY THE RIGHT NUTRIENTS IN THE RIGHT AMOUNTS THE BABY NEEDS AT DIFFERENT STAGES	IF IT DOES NOT COME EASILY TO THE MOTHER IT CAN CONTRIBUTE TO POSTNATAL DEPRESSION
	CONTAINS ANTIBODIES WHICH HELP THE BABY PREVENT INFECTION	RESPONSIBILITY FOR FEEDING THE BABY IS SOLELY WITH THE MOTHER
	HELPS DEVELOP A BOND BETWEEN MOTHER AND BABY	
	IS FREE	
BOTTLE FEEDING	ALLOWS THE FATHER TO BOND WITH THE BABY	RISK OF INFECTION IS INCREASED IF BOTTLES AND EQUIPMENT ARE NOT PROPERLY STERILISED
	ALLOWS THE MOTHER MORE FREEDOM AS SHE IS NOT SOLELY RESPONSIBLE FOR FEEDING THE BABY	IS EXPENSIVE



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16.6 HUMAN SEX HORMONES

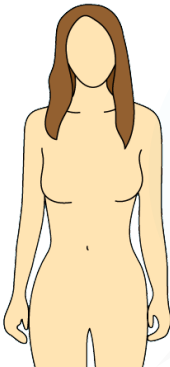
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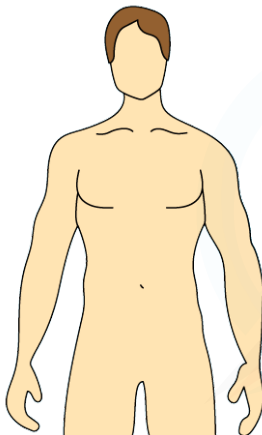
Secondary Sexual Characteristics

- Primary sexual characteristics are present during development in the uterus and are the differences in reproductive organs etc between males and females
- Secondary sexual characteristics are the **changes that occur during puberty** as children become adolescents
- They are controlled by the release of **hormones** – **oestrogen** in girls and **testosterone** in boys

Female secondary sexual characteristics

FEMALE	EFFECTS OF OESTROGEN
	BREASTS DEVELOP
	BODY HAIR GROWS
	MENSTRUAL CYCLE BEGINS
	HIPS GET WIDER

Male secondary sexual characteristics

MALE	EFFECTS OF TESTOSTERONE
	GROWTH OF PENIS & TESTES
	GROWTH OF FACIAL & BODY HAIR
	MUSCLES DEVELOP
	VOICE BREAKS
	TESTES START TO PRODUCE SPERM

- Some changes occur to both boys and girls, including **growth of sexual organs** and **growth of body hair**
- **Emotional changes** also occur due to the increased levels of hormones in the body
- These include more interest in sex and increased mood swings



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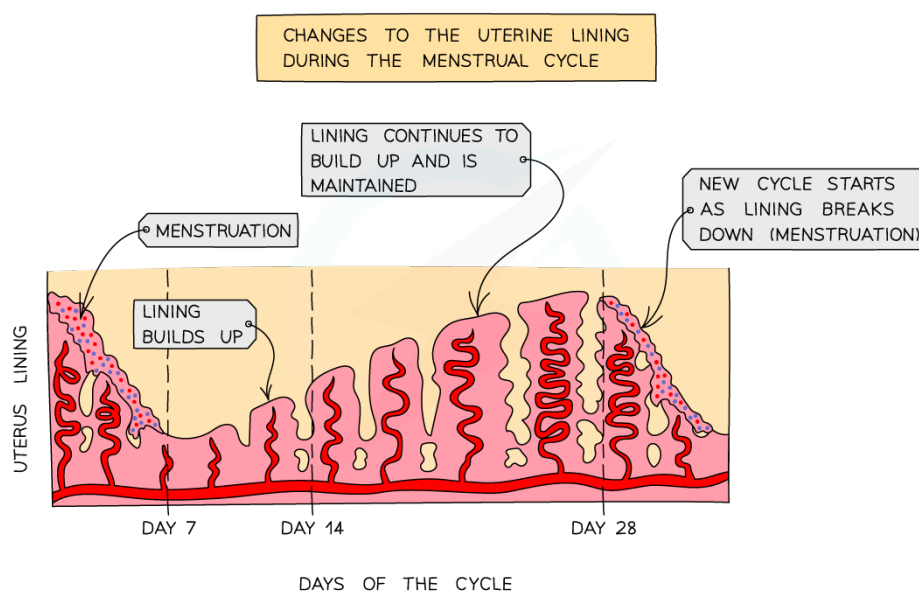
16.6 HUMAN SEX HORMONES cont...

YOUR NOTES



The Menstrual Cycle

- Starts in early adolescence in girls (around age 12) and is controlled by **hormones**
- The average menstrual cycle is **28 days** long
- **Ovulation** (the release of an egg) occurs about **halfway** through the cycle (day **14**) and the egg then travels down the oviduct to the uterus
- Failure to fertilise the egg causes **menstruation** (commonly called a period) to occur – this is caused by the **breakdown of the thickened lining of the uterus**
- Menstruation lasts around **5 – 7 days** and signals the beginning of the next cycle
- After menstruation finishes, the lining of the uterus starts to thicken again in preparation for possible implantation in the next cycle



Changes in the lining of the uterus during the menstrual cycle



EXTENDED ONLY

Hormonal Control of the Menstrual Cycle

- The menstrual cycle is controlled by hormones released from the **ovary** and the **pituitary gland** in the brain



16 REPRODUCTION

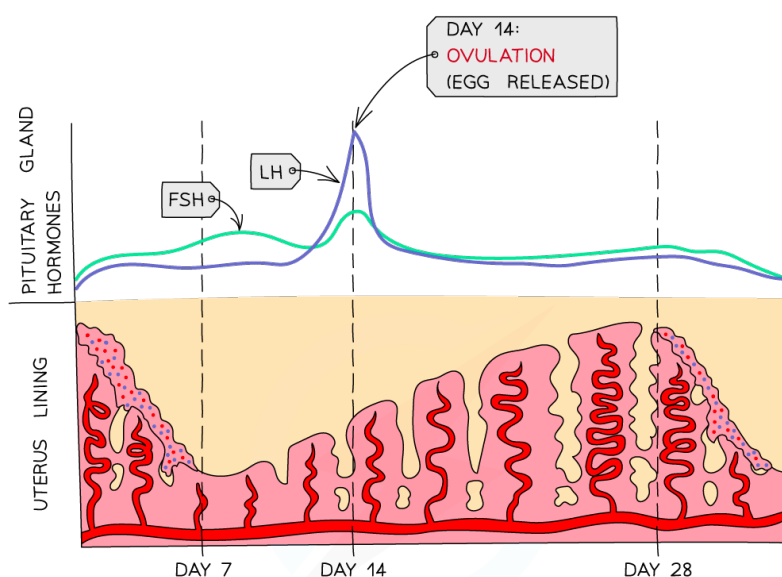
16.6 HUMAN SEX HORMONES cont...

YOUR NOTES



EXTENDED ONLY cont...

FSH and LH



FSH

- STIMULATES EGG MATURATION IN THE FOLLICLES OF THE OVARY
- STIMULATES FOLLICLES IN THE OVARIES TO SECRETE OESTROGEN

LH

- AT ITS PEAK STIMULATES OVULATION (RELEASE OF EGG INTO OVIDUCT)
- RESULTS IN THE FORMATION OF A CORPUS LUTEUM

Changes in the levels of the pituitary hormones FSH and LH in the blood during the menstrual cycle

- **FSH** (follicle stimulating hormone) is released by the **pituitary gland** and causes an **egg to start maturing** in the ovary
- It also **stimulates the ovaries** to start releasing **oestrogen**
- The **pituitary gland** is stimulated to release **luteinising hormone (LH)** when **oestrogen** levels have reached their peak
- **LH** causes **ovulation to occur** and also **stimulates the ovary** to produce **progesterone**



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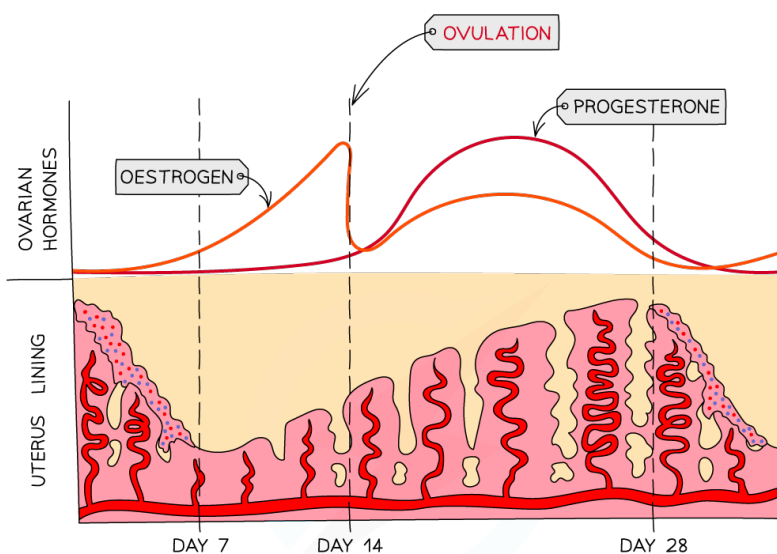
16.6 HUMAN SEX HORMONES cont...

YOUR NOTES



EXTENDED ONLY cont...

The roles of oestrogen and progesterone



OESTROGEN

- STIMULATES THE UTERUS TO DEVELOP A LINING (TO REPLACE THE LINING LOST DURING MENSTRUATION)
- POST-OVULATION, INHIBITS FSH AND LH PRODUCTION IN THE PITUITARY GLAND

PROGESTERONE

- MAINTAINS AND THICKENS LINING OF THE UTERUS
- INHIBITS FSH AND LH PRODUCTION
- IF FERTILISATION DOESN'T OCCUR, LEVELS DROP AND MENSTRUATION OCCURS.

Changes in the levels of oestrogen and progesterone in the blood during the menstrual cycle

- **Oestrogen** levels rise from day 1 to peak just before **day 14**
- This causes the **uterine wall to start thickening** and the **egg to mature**
- The peak in oestrogen occurs just before the egg is released
- **Progesterone** stays low from day 1 – 14 and starts to rise once ovulation has occurred
- The increasing levels **cause the uterine lining to thicken further**; a fall in progesterone levels causes the uterine lining to break down (**menstruation / 'period'**)



16 REPRODUCTION

16.6 HUMAN SEX HORMONES cont...

YOUR NOTES



EXTENDED ONLY cont...

Interaction between all four of the menstrual cycle hormones

- The pituitary gland produces **FSH** which stimulates the development of a **follicle** in the ovary
- An egg develops inside the follicle and the follicle produces the hormone **oestrogen**
- Oestrogen causes **growth and repair of the lining of the uterus wall** and inhibits production of **FSH**
- When oestrogen rises to a high enough level it stimulates the release of **LH** from the pituitary gland which causes **ovulation** (usually around day 14 of the cycle)
- The follicle becomes the **corpus luteum** and starts producing **progesterone**
- Progesterone **maintains the uterus lining** (the thickness of the uterus wall)
- If the ovum is not fertilised, the corpus luteum breaks down and progesterone levels drop
- This causes **menstruation**, where the uterus lining breaks down and is removed through the vagina – commonly known as having a period
- If pregnancy does occur the corpus luteum **continues to produce progesterone**, preventing the uterus lining from breaking down and **aborting** the pregnancy
- It does this until the **placenta** has developed, at which point it starts secreting progesterone and **continues to do so throughout the pregnancy**

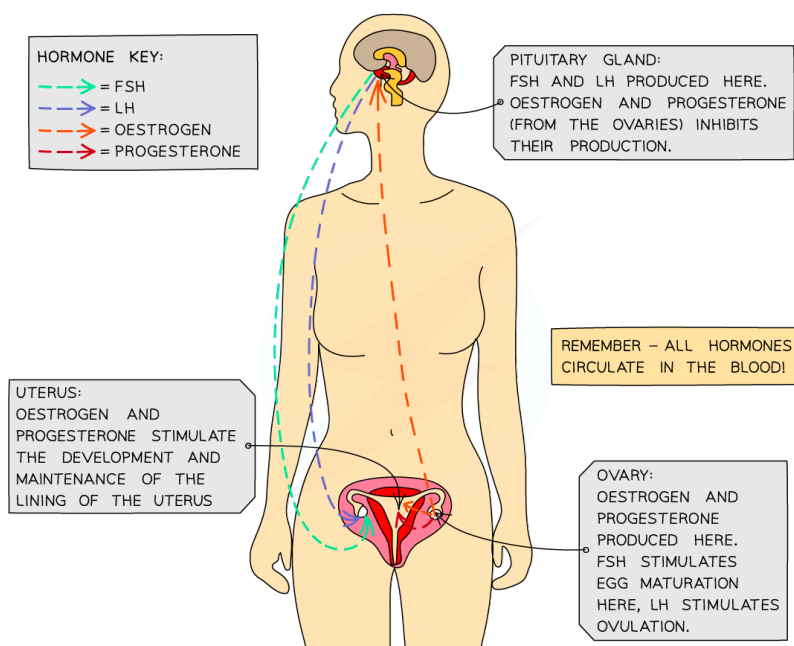


Diagram showing where hormones involved in the menstrual cycle are made and act



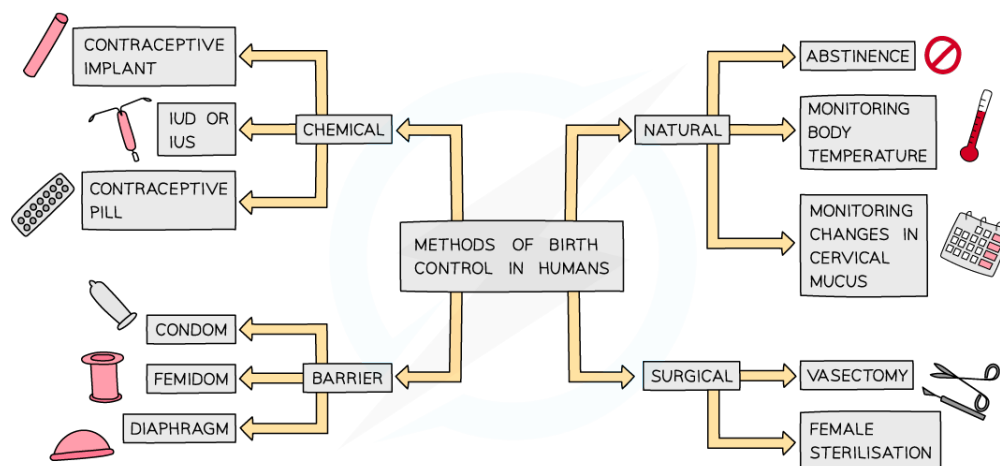
16 REPRODUCTION

16.7 CONTRACEPTION & FERTILITY

YOUR NOTES



Types of Birth Control



- Birth control methods are important in **keeping family sizes small** and in **limiting the increase in human population**
- Humans can use mechanical, chemical, surgical and natural contraceptive methods to prevent a pregnancy
- Some birth control methods also give protection from **sexually transmitted infections**

Natural

- **Abstinence**
 - avoiding sexual intercourse completely
- **Rhythm method**
 - avoiding sexual intercourse during the fertile period of the menstrual cycle when ovulation occurs
 - the exact time ovulation happens can be worked out by monitoring body temperature and quality of cervical mucus
 - this is the least reliable method of birth control

Chemical

- **IUD / IUS**
 - an intrauterine device or intrauterine system is a small device fitted inside the uterus by a doctor or nurse
 - it releases sex hormones which thicken the mucus produced in the cervix, making it difficult for sperm to swim into the uterus



16 REPRODUCTION

16.7 CONTRACEPTION & FERTILITY cont...

YOUR NOTES



- it also thins the lining of the uterus, making it more difficult for a fertilised egg to implant
- an IUD also interferes with passage of sperm through the uterus, in which way it is acting as a barrier method of birth control
- **Contraceptive pill, implant, injection**
 - may contain just progesterone or a mixture of progesterone and oestrogen
 - very effective when taken regularly
 - the hormones can also be delivered from a small skin implant or an injection, both of which last several months and increase the effectiveness as they remove the risk of forgetting to take a pill regularly

Barrier

- These all work by preventing sperm from reaching the egg
- **Condom**
 - latex sheath worn over the penis
 - prevents sperm entering the vagina as ejaculate remains in condom
 - also protects against STIs
- **Femidom**
 - latex sheath inserted into the vagina
 - prevents entry of sperm into the vagina
- **Diaphragm**
 - a rubber cap that fits over the entrance to the cervix
 - prevents entry of sperm into uterus
 - often used with a spermicide (cream which kills sperm)

Surgical

- **Vasectomy**
 - the sperm ducts are cut, meaning that no sperm is present in the semen when ejaculation occurs
 - very effective but difficult to reverse
- **Female sterilisation (tubal ligation)**
 - the oviducts are cut or tied off, preventing eggs from reaching the uterus or sperm from reaching the eggs
 - very effective but difficult to reverse



16 REPRODUCTION

16.7 CONTRACEPTION & FERTILITY

YOUR NOTES



EXTENDED ONLY

How Do Contraceptive Hormones in The Pill Work?

- They work by mimicking some of the hormone levels during pregnancy
- By raising the levels of progesterone and oestrogen, the uterus lining is maintained and development of another egg cell is prevented
- This means that sex at any time of the month cannot cause pregnancy as no egg is released to be fertilised

Fertility Treatments

- In situations where couples find it difficult to conceive, fertility treatments can improve their chances
- There are several different options, depending on what the fertility issue is

Artificial Insemination (AI)

- If the male is **not producing healthy sperm**, donor sperm can be used
- The sperm are placed into the female's vagina at the fertile point in her menstrual cycle
- **Social issues to consider include:**
 - The male must be able to cope with the fact that the child is not biologically his
 - Sperm donors may father multiple children who are not able to know their parentage

Fertility Drugs

- This method is used when the **female is not producing enough eggs**
- Hormones, including FSH, are given to her to **stimulate egg production**
- **Social issues to consider include:**
 - Several eggs can be released at once so this increases the chance of multiple births (twins or triplets etc)



16 REPRODUCTION

16.7 CONTRACEPTION & FERTILITY cont...

YOUR NOTES



EXTENDED ONLY

In Vitro Fertilisation (IVF)

- If the female cannot conceive naturally even after taking fertility drugs, or if there are issues with both male and female fertility in a couple, **IVF** can be used
- This involves fertility drugs being given to the female to stimulate egg production before they are **harvested from the ovary**
- The eggs are then **inseminated in a petri dish** using sperm from the male ('in vitro' means 'in glass') and, **once embryos have formed**, they are **placed back into the uterus** of the female
- **Several embryos are implanted** to increase the chance of one developing further
- **Social issues to consider include:**
 - IVF is relatively expensive and not all couples can afford it
 - As several embryos are implanted, the risk of multiple births is quite high
 - Some women use IVF to get pregnant at a later age than they would be able to conceive naturally

16.8 SEXUALLY TRANSMITTED INFECTIONS

STIs & HIV / AIDS

- Unprotected sexual intercourse can lead to the transfer of pathogens via exchange of body fluids
- Infections passed on in this way are known as sexually transmitted infections (STIs)
- An example of an STI is HIV, the virus that usually leads to the development of acquired immunodeficiency disease (AIDS)
- HIV can also be spread via sharing needles with an infected person, blood transfusions with infected blood and from mother to fetus through the placenta and mother to baby via breastfeeding



16 REPRODUCTION

16.8 SEXUALLY TRANSMITTED INFECTIONS cont...

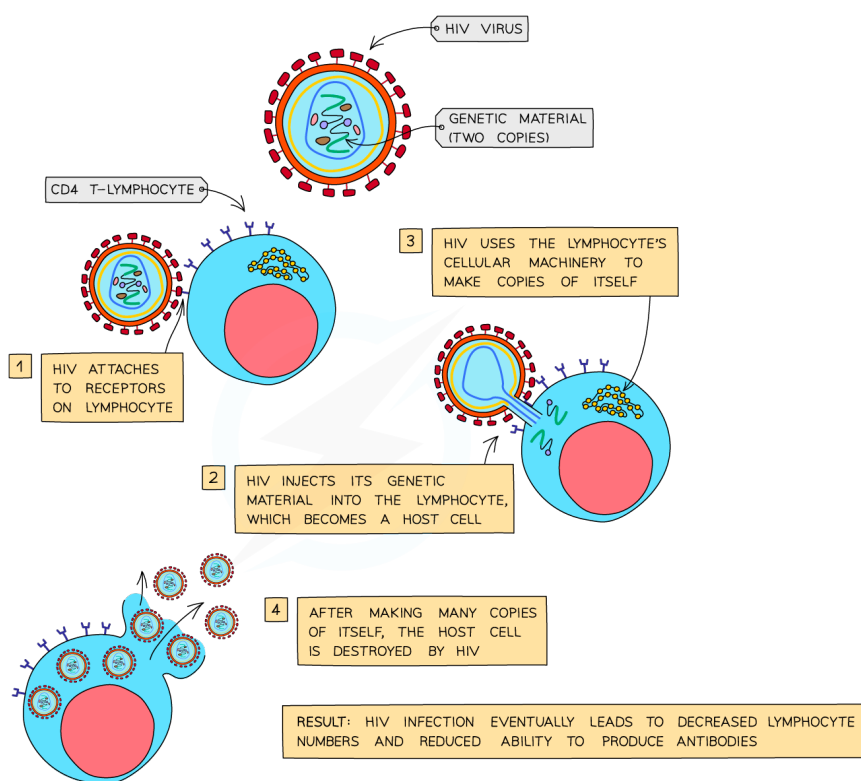
YOUR NOTES



EXTENDED ONLY

How HIV Affects the Immune System

- Immediately after infection, people often suffer **mild flu-like symptoms**
- These symptoms pass and for a period of time **infected people might not know they are infected**
- The virus infects a certain type of **lymphocyte** of the body's immune system
- Normally lymphocytes **seek out and destroy pathogens** that enter the body, producing antibodies that attach to pathogens, enhancing phagocytic activity
- However HIV avoids being recognised and destroyed by lymphocytes by repeatedly **changing its protein coat**
- It then infects a certain type of lymphocyte and **uses the cells' machinery to multiply**
- This **reduces the number of lymphocytes** of the immune system, and also **the number of antibodies** that can be made
- This decreases the body's ability to fight off infections, eventually leading to **AIDS (Acquired immunodeficiency)**



How HIV affects lymphocytes



16 REPRODUCTION

16.8 SEXUALLY TRANSMITTED INFECTIONS cont...

YOUR NOTES



Controlling the Spread of STIs

- The spread of STIs such as HIV are best controlled by:
 - **Limiting the number of sexual partners** an individual has
 - **Not having unprotected sex**, but making sure to always use a **condom**
 - **Getting tested** if unprotected sex or sex with multiple partners has occurred
 - Raising awareness by **education programmes**

> NOW TRY SOME EXAM QUESTIONS



16 REPRODUCTION

EXAM QUESTIONS

YOUR NOTES



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QUESTION 1

During birth the following processes happen

- 1 Breaking of the amniotic sac
- 2 Contraction of the muscles in the uterine wall
- 3 Cutting of the umbilical cord

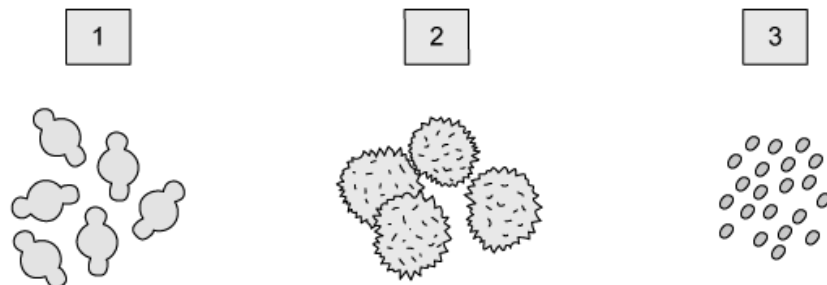
In which order do these processes usually occur?

- A** 2 → 3 → 1 **B** 2 → 1 → 3 **C** 3 → 1 → 2 **D** 3 → 2 → 1

?

QUESTION 2

The diagrams show 3 different species of pollen grains as they appear under a microscope. The diagrams are all to the same scale.



Which pollen grains are involved in insect pollination?

- A** 1 and 2
B 2 and 3
C 2 only
D all of them



16 REPRODUCTION

EXAM QUESTIONS cont...

YOUR NOTES



?

QUESTION 3

The table below shows some advantages and disadvantages of sexual and asexual reproduction.

Which row is correct?

	advantage of sexual reproduction	advantage of asexual reproduction
A	less population growth	only one parent needed
B	more energy efficient	gametes can be transferred by the environment
C	no transfer of gametes	does not compete with parent for nutrients
D	more genetic variation	faster

?

QUESTION 4

Students were discussing how to remember the difference between two groups of cells, 'many, minute and motile' or 'few, fat and fixed'

What group of cells were they referring to?

- A** male and female gametes
- B** animal and plant cells
- C** red and white blood cells
- D** xylem and phloem cells



16 REPRODUCTION

EXAM QUESTIONS cont...

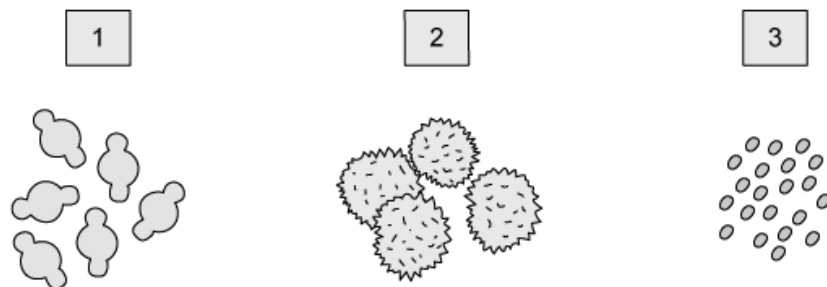
YOUR NOTES



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QUESTION 5

The diagram below shows a flower.



Which of these processes have taken place?

	pollination	fertilisation
A	no	no
B	no	yes
C	yes	yes
D	yes	no

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